

# JY Liu

Solutions Architect | Technical Lead

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## Summary

- Solutions architect and technical lead with 12+ years shipping web, mobile, cloud, and integration-heavy platforms across fintech, SaaS, banking, consular, and enterprise environments.
- Hands-on across frontend, backend, APIs, DevOps, and cloud infrastructure — consistently taking products from discovery and architecture through production delivery.
- Leverages AI tooling pragmatically across the SDLC — analysis, implementation, prototyping, and quality assurance — while building AI-native products end-to-end and maintaining production-grade technical rigor.

## Experience

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| <b>Vailley AI Solution Pte. Ltd.</b> , Senior Web Developer (Technical Lead)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Remote (Singapore)<br>Jun 2025 – Present         |
| <ul style="list-style-type: none"><li>• Leading cross-functional delivery for VFS Global's visa and consular platforms — combining solution architecture with hands-on implementation across the application stack, integration layer, and platform.</li><li>• Delivered a Consular Dashboard with RBAC and ABAC controls, an ADR portal with structured document submission, ETL-backed operational visibility, a digital corporate visa platform for agents, and SAP Concur to OneVasco integration for VFS Global.</li></ul>                                          |                                                  |
| <b>Shift Markets</b> , Senior Software Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Remote (NY, US)<br>Jan 2023 – Present            |
| <ul style="list-style-type: none"><li>• Lead engineer for CXM Direct (worldwide multi-asset trading platform) and CRM initiatives, owning backend delivery with ThinkPHP and frontend execution in TypeScript.</li><li>• Engineered a Unified Payment Gateway integrating 20+ global Payment Service Providers (PSPs), unifying fragmented payment workflows into a single scalable infrastructure.</li><li>• Strengthened system reliability and delivery velocity through architectural cleanup, perf tuning, and production-oriented engineering practices.</li></ul> |                                                  |
| <b>InternetBrands</b> , Senior Web Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Remote (California, US)<br>Apr 2023 – Sept 2024  |
| <ul style="list-style-type: none"><li>• Maintained and improved the Dentrix Hub Forms (Online Form for Dentrix &amp; HOP) Laravel backend — patient form management platform for dental practices, focused on reliability, performance, and scale.</li><li>• Introduced PHP linting and unit testing standards, increasing code coverage.</li><li>• Managed Kubernetes deployment with Helm and Rancher for the backend.</li></ul>                                                                                                                                       |                                                  |
| <b>MaiDaLv</b> , Senior Full Stack Developer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Remote (Guangzhou, China)<br>Feb 2025 – Apr 2025 |
| <ul style="list-style-type: none"><li>• Built Spring Boot, UniApp, Vue.js, and Python-based product workflows covering subs, WeChat integration, and IP-risk scan.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                             |                                                  |
| <b>Cake Experiential Communications Asia</b> , Solutions Architect                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Remote (KL, Malaysia)<br>Apr 2021 – May 2023     |
| <ul style="list-style-type: none"><li>• Led end-to-end architecture and delivery for Vue.js and Laravel platforms used in regional enterprise and campaign engagements.</li><li>• Introduced engineering standards — TDD, API governance, and backend scaling practices — for client-critical systems.</li></ul>                                                                                                                                                                                                                                                         |                                                  |
| <b>Glints</b> , Web Software Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Remote (Singapore)<br>Jan 2022 – Dec 2022        |
| <ul style="list-style-type: none"><li>• Built full-stack recruiter tooling for Superpowered using Next.js and Koa.js, with emphasis on workflow efficiency and usable internal product design.</li><li>• Strengthened dashboard responsiveness and release quality through improved testing and deployment automation.</li></ul>                                                                                                                                                                                                                                         |                                                  |

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| <b>InnoLab Global</b> , Senior Information Technology Consultant <ul style="list-style-type: none"> <li>Delivered cross-platform products across Angular, Ionic, Flutter, Firebase, and AWS while automating cloud delivery workflows and mentoring junior developers.</li> <li>Shipped the Forest City Super App on a Turborepo + Next.js + Prisma + PostgreSQL + Auth.js stack covering community, events, payments, partner portal, and admin console features for residents and partners.</li> <li>Built Aegis Shooting Academy (ASA) digital platform end-to-end with NestJS + Prisma backend, Next.js admin dashboard, and React Native/Expo mobile application.</li> <li>Delivered a corporate website and WooCommerce product catalogue for 3V Malaysia (furniture manufacturer) using Docker Compose with a premium Apple-inspired theme.</li> <li>Architected and led delivery for AvaEra's compensation platform — modernized legacy stack into pnpm + Turborepo monorepo with Next.js, Prisma, PostgreSQL, and a 15-service reward engine covering direct sponsor, team bonus, pool sharing, profit distribution, and merchant commissions.</li> <li>Built and shipped HappyAI for Gembira Bina Sdn Bhd, an AI-powered tender document extraction and validation platform — Next.js + FastAPI + SQLite with Groq LLM primary / Ollama fallback, confidence-badged review UI, and deterministic batch ingest pipeline.</li> </ul> | Remote (KL, Malaysia)<br>Sept 2018 – Present  |
| <b>Supahands</b> , Full Stack Developer <ul style="list-style-type: none"> <li>Built full-stack features for Supahands' distributed-team product.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Remote (KL, Malaysia)<br>Feb 2018 – Jun 2018  |
| <b>The Consociation LLP</b> , Information Technology Consultant <ul style="list-style-type: none"> <li>Led full-stack delivery for products including iHeartCity and Swiiipe using Angular, NestJS, GraphQL, and Firebase with AWS-based DevOps automation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Remote (Singapore)<br>Sept 2016 – Apr 2020    |
| <b>Buddy Holdings Pte Ltd</b> , Senior Information Technology Consultant <ul style="list-style-type: none"> <li>Architected and led delivery of the Buddy System Asia MLM operations platform — Angular + Laravel + MySQL + Kafka covering hierarchy management, complex point calculations, point redemption, inventory tracking, and family tree updates at scale.</li> <li>Built agent-facing UI for real-time tracking of progress, points, and family trees.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Remote (Singapore)<br>Sept 2016 – Jan 2024    |
| <b>Fuji Xerox Pte Ltd</b> , Embedded Software Engineer <ul style="list-style-type: none"> <li>Developed embedded software for printer/scanner hardware platforms with C/C++.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Singapore<br>Feb 2016 – May 2016              |
| <b>Netccentric Pte Ltd</b> , Software Developer <ul style="list-style-type: none"> <li>Built full-stack web features for digital ad campaigns.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Singapore<br>Oct 2015 – Dec 2015              |
| <b>Nexstream Sdn Bhd</b> , Hybrid App Developer <ul style="list-style-type: none"> <li>Built hybrid mobile apps for iOS and Android using web technologies.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Kuala Lumpur, Malaysia<br>Jun 2014 – Jul 2015 |

## Selected Projects

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| <b>Unified Payment Gateway</b><br>Specialized payment infrastructure unifying 20+ global Payment Service Providers (PSPs) into a single, high-concurrency API layer. <ul style="list-style-type: none"> <li>Architected and delivered a unified integration layer for diverse PSPs, handling complex transaction workflows, reconciliation, and automated settlement.</li> <li>Secured financial operations through strict audit logging, sensitive data isolation, and robust error-recovery mechanisms.</li> </ul> | Jan 2023 – Present |
| <b>VFS Global Consular Platform</b><br>Mission-critical visa and consular services platform delivering digital transformation for global diplomatic missions. <ul style="list-style-type: none"> <li>Delivered a high-security Consular Dashboard with granular RBAC/ABAC and integrated SAP Concur for automated expense and travel management.</li> <li>Built a digital corporate visa platform enabling agents to manage large-scale applications with real-time tracking and ETL-backed analytics.</li> </ul>    | Jun 2025 – Present |

**Superpowered**  
Talent platform delivered with scalable web architecture, backend APIs, GraphQL, PostgreSQL, and event-driven workflows.

- Built a responsive Next.js and Koa.js application with GraphQL integration, backend optimization, and scalable recruiter tools.

Jan 2022 – Dec 2022

**Dentrix Hub Forms (Online Form for Dentrix & HOP)**  
Dental practice patient form management platform — Laravel backend with Kubernetes deployment, PHP linting, and unit testing standards.

- Maintained and improved the Laravel backend, strengthening response times and concurrency handling for patient form workflows.
- Introduced PHP linting and unit testing, increasing code coverage.
- Managed Kubernetes deployment with Helm and Rancher for scalable, reliable ops.

Apr 2023 – Sept 2024

Education

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|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>Master of Business Administration University of Malaya</b> , Business Analytics<br>(Expected Oct 2026) | Kuala Lumpur, Malaysia<br>Oct 2023 – Oct 2026 |
| <b>Master of Computer Science      University of Malaya</b> , Mobile Cloud Computing                      | Kuala Lumpur, Malaysia<br>Jan 2011 – Jan 2013 |
| <b>Bachelor of Computer Science      Universiti Sains Malaysia</b> , Software Engineering                 | Malaysia<br>Jan 2008 – Jan 2011               |

Skills

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**Full Stack Engineering:** TypeScript, JavaScript, PHP, Python, Java, Laravel, ThinkPHP, React, Next.js, Angular, Vue.js, NestJS, Node.js, Spring Boot, Flutter, Ionic

**Architecture, Cloud & DevOps:** Solution architecture, AWS, GCP, Firebase, Docker, Kubernetes, Helm, Rancher, CI/CD, deployment automation, integration-heavy platform delivery

**Backend, Data & Integration:** PostgreSQL, MySQL, Redis, Apache Kafka, GraphQL, REST APIs, microservices, ETL workflows, event-driven and serverless systems

**AI-Enabled Delivery & Multi-Agent Systems:** Orchestrating LLM-based agents, RAG pipelines, prompt engineering, and custom GPT development to accelerate product delivery.

Languages

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**English, Malay, Cantonese:** Professional

**Mandarin:** Native

**Japanese, Korean:** Beginner